THE KERNEL, THE RECEIPTS AND THE TWO-LAYER CLOSURE

THE MASTER v1.1.0_CC · PARTS, NAMESPACES, THEOREMS, CONES

PART	NAMESPACE	THEOREMS	CONES	WHAT IT PROVES
0	`BSDHard`	**101**	**∅ every one**	The kernel of record, verbatim: the seat, the order, the act, parity, low order, Ω, F_2
A	`BSD.NORMALFORM`	**25**	**∅ every one**	The normal form: the record carries the value; the value is the transfer
B	`BSD.HOST`	**16**	**∅ every one**	The host and its binding; the sign fixes the parity; curves of every order
C	`BSD.ROWSPLIT`	**7**	**∅ every one**	The cut-agnostic division, carried verbatim from the Riemann master
C′	`BSD.REALPART`	**9**	**∅ every one**	The division read at order one, the low-order witness the real part
D	`BSD.WALLS`	**4**	**∅ every one**	The sign record normal form; the walls in one theorem
E	`BSD.RESIDUE`	**7**	**∅ every one**	The one implication, printed as a hypothesis and assumed nowhere
F	`BSD.SUPPLY`	**12**	**∅ every one**	The bit delivered twice; the witnesses typed; the one refuter
G	`BSD.CLOSURE`	**5**	**∅ every one**	The question closed to one bit
W	`BSD.GATES`	**15**	**∅ every one**	The twelve formal gates and the four refusals
X	`BSD.ESCAPE`	**9**	**∅ every one**	Nothing escapes; our way or no way; the capstone
T	`BSD.TERMINAL`	**4**	**∅ every one**	The open framing refuted; the ten steps; the final form
Bind	`BSDMathlib`	**18**	**∅ · std**	Mathlib's curves and rank; 37a1 and 389a1 computed by the library's group law

RECEIPTS

Master	`BSD_Normal_Form_Paper_v1_1_0_CC.lean`
SHA-256	`66dff84e8bc25f950b48bf39909ccf9d0be4cabd9f1c3ee9721ac136e83a7112`
Build	`Lean 4.19.0 core · no import · exit 0, no message · 17 s`
Cones	`214 theorems, 214 pinned by #guard_msgs, every cone empty`
Part 0, the kernel of record	`BSD_Hardened.lean · d52b7a9e5b4f9be4…c8be`
Mathlib binding	`BSD_Mathlib_Binding_v1_1_0_CC.lean · 18 theorems · Mathlib v4.19.0`
Binding SHA-256	`c9e2c78d0644cd687f133236437557100942 4d64a5d74da299db53388fd4dbf8`
Appendix R	`TOE_Zero · Triaxial_Actuation · Root_Grade_Ledger · 65 theorems on no axiom`

THE TWO-LAYER CLOSURE

Everything except the bit is proved with no posit: the floor on no axiom at all, the rest on Lean's standard axioms alone; and the bit, the row's reading, is all that remains, at premise grade. In this paper every theorem of the master stands on no axiom at all, and only the Mathlib binding reaches the standard axioms.

READING theorem on no axiom at all · binding theorem on Lean's standard axioms · definition or structure — names shown by their last component